

Ibrahim Ahmad Siddiqi

Backend & AI Engineer | LLM Systems & Full-Stack Developer

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EXECUTIVE SUMMARY

Backend and AI Engineer with hands-on production experience at **Systems Limited** and a research lab. Specializing in intelligent backend architectures, RAG pipelines, multi-agent orchestration, and production-grade REST APIs. Proficient in **Python, Node.js/Express.js, React, FastAPI**, with extensive experience integrating modern LLM models (Groq, Gemini, Cohere) and vector databases (FAISS, Qdrant, ChromaDB). Completing BS in Software Engineering at **FAST-NUCES Islamabad (2026)**.

TECHNICAL SKILL MATRIX

AI & LLM Systems

RAG Pipelines Multi-Agent Orchestration LangGraph
Groq (Llama3) Gemini API Cohere
Prompt Engineering FAISS Qdrant ChromaDB
Sentence Transformers NLP

ML & Deep Learning

Python TensorFlow PyTorch CNN LSTM
BERT RoBERTa Scikit-learn NumPy Streamlit

Backend Development

Python (FastAPI, Flask) Node.js Express.js
RESTful API Design JWT Authentication Rate Limiting
PostgreSQL MongoDB SQLite

Frontend & DevOps

React Next.js JavaScript PyQt6 Docker
Kubernetes AWS Rekognition Git / GitHub
Postman

PROFESSIONAL WORK EXPERIENCE

Full-Stack Python Intern • Research Lab

Remote | 06/2026 – 08/2026

- Engineered **EcoTrack** — a PyQt6 cross-platform desktop application monitoring active system processes and calculating real-time CO₂ emissions using IEA grid carbon-intensity factors.
- Implemented historical analytics dashboard and data export functionality for emissions tracking over time.
- Delivered across iterative releases, incorporating review feedback across UI, database error handling (SQLite), and packaging to ship a stable, production-ready build.

Python

PyQt6

SQLite

IEA API

Data Analytics

Software Engineering Intern • Systems Limited

Islamabad, Pakistan | 06/2025 – 08/2025

- Engineered a multi-layer validation engine (AI semantic checks, schema integrity, data normalization) across an Express.js, SQLite, and React stack.
- Built **SPECmate** — an automated LLM-powered SRS artifact generator (Llama3-70B via Groq) auto-producing test cases, use cases, and functional modules from raw requirements.
- Collaborated with cross-functional digital consulting teams delivering enterprise-grade solutions using agile development practices.

React

Express.js

SQLite

Node.js

LLMs (Groq / Llama3)

RESTful APIs

FEATURED PROJECTS PORTFOLIO

1. Multi-Agent AI Orchestration System

09/2025 – 11/2025

Gemini-powered orchestration for a 13-agent educational AI system with two-tier memory and intent dispatch.

- Co-engineered Gemini-powered orchestration for a 13-agent educational AI system: single-call intent dispatch with confidence scoring and conversational clarification, cutting latency and cost.
- Added a two-tier memory layer (short-term session recall, per-agent long-term store) for context continuity across multi-turn conversations.
- Implemented LangGraph state machine for agent routing, tool calling, and graceful fallback handling.

FastAPI LangGraph Gemini API ChromaDB Python

2. SEOmation — AI SEO Content Platform

08/2025 – 02/2026

3-service content automation platform with event-driven scheduler, multi-engine research pipeline, and LLM content generator scoring 85+ on SEO tools.

- Built a 3-service architecture (Express.js/PostgreSQL, Python AI service, React frontend) with automatic repair passes for malformed responses.
- Event-driven Node.js scheduler with retry/backoff and missed-job recovery for reliable autonomous publishing.
- Integrated Qdrant vector store and Cohere embeddings to retrieve real-time web context for RAG generation.

Express.js Flask Groq PostgreSQL Qdrant Cohere React

3. Detecting AI-Generated vs. Human-Written Text

2025

Transformer-based binary classifier (BERT & RoBERTa) with critical analysis of overfitting vs real-world generalization.

- Implemented and compared Logistic Regression, BERT (110M params), and RoBERTa (125M params) on HC3 and Kaggle datasets (~10,000 samples).
- Achieved 99.91% validation accuracy with RoBERTa (AUC-ROC 0.9999). Aggressive regularization produced 97.26% robust generalization accuracy.

Python BERT RoBERTa TensorFlow NLP Hugging Face

4. RAG-based FYP Handbook Assistant

11/2025

Performance retrieval pipeline using Sentence Transformers and FAISS for intelligent handbook queries.

- Processed 50+ handbook pages into contextual chunks with sub-400ms query latency.
- Intelligent chunking with metadata retention providing context-grounded answers with automated page citations in Streamlit.

Python Sentence-BERT FAISS Streamlit

5. Sign Language Digit Recognition

11/2025

Deep CNN for classifying sign language digit gestures across 10 classes.

- Designed and trained a deep CNN to classify 64×64 sign-language digit images with high multi-class accuracy.
- Evaluated with confusion matrices, precision-recall, and AUC-ROC curves.

CNN TensorFlow Python Deep Learning

6. Collaborative Recipe & Meal Planner

05/2025

Full-stack meal planning platform with collaborative features and automated scheduling.

- Express.js/MongoDB meal platform with JWT auth serving 400+ recipes & supporting 80+ concurrent users.
- React SPA containerized with Docker and deployed on Kubernetes via CI/CD pipeline.

React Node.js Express.js MongoDB Docker Kubernetes

7. Image Moderation App

05/2025

Role-based image moderation platform with AI-powered content detection using AWS Rekognition.

- FastAPI/MongoDB app with JWT authentication & role separation; AWS Rekognition integration achieving 95% accuracy.

FastAPI MongoDB AWS Rekognition Docker Python

8. University Timetable Scheduler

04/2025 – 05/2025

AI scheduling system using Constraint Satisfaction Problems (CSP) & Genetic Algorithms.

- Engineered CSP/GA scheduler for 1500+ students across 120+ courses, achieving 90% efficiency with MRV heuristics.
- Reduced computation time by 50% using GA crossover/mutation (25-second execution).

Python CSP Genetic Algorithms NumPy

9. Movie Recommending System

11/2024 – 12/2024

Collaborative filtering recommendation engine with user authentication and preference tracking.

- Express.js/MongoDB recommendation engine with 90% accuracy for 150+ active users.

Express.js MongoDB Node.js JWT

10. Next-Word Prediction System

11/2025

LSTM-based word prediction model trained on Shakespeare's plays with interactive UI.

- Word-level LSTM model predicting next words in real-time as users type partial sentences.

LSTM NLP Python TensorFlow

11. Rate-My-Professor RAG

08/2024

Professor review system with AI-powered response generation using custom RAG pipeline.

- Flask API gateway & Sentence Transformers processing 100+ professor reviews with sub-400ms latency.

Python

Next.js

RAG

Pinecone

Flask

EDUCATION & CERTIFICATIONS



Education

National University of Computer & Emerging Sciences (FAST-NUCES)

Islamabad, Pakistan

BS in Software Engineering

Graduation: June 2026

Relevant Coursework:

Applied AI

Cloud Computing

Web Engineering

DevOps

Software Architecture

Algorithms & Data Structures

Database Systems



Certifications & Leadership

McKinsey Forward Program

McKinsey & Company (2026)

AWS Graduate – Cloud Foundations

Amazon Web Services (2026)

AWS Graduate – Cloud Web App Builder

Amazon Web Services (2026)

AWS Cloud Quest: Cloud Practitioner

Amazon Web Services (2026)

Intro to Generative AI for Devs

DeepLearning.AI (2025)

Campus Leadership:

- Google Developer Student Club — Technical Workshop Team
- FAST Software Engineering Society — Media Head
- NASCON — Management Officer